

TEN PROBLEMS You Can SOLVE With A MULTIMETER



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We live in a pool of electronic instruments and gadgets that we use extensively in our daily lives. In case of minor issues with these instruments, you do not have to be an electronics expert or call an electrician to resolve it. With a simple multimeter, you can be an expert yourself.

Multimeters, also known as daily-life meters, are excellent tools that can measure current, voltage and resistance. With the values of these three parameters in hand, most electrical problems can be easily fixed. These multimeters can be used for continuity and diode checks, which helps in solving many issues. Overall, these are simple tools but save a lot of money and time, and make work a lot easier.

In the USA, electrical outlets provide 110 volts, and heavy-duty appliance circuits (electric stoves, clothes dryers, etc) use 230 volts. In many other parts of the world, including India, standard voltage for household electrical systems is 230 to 240 volts.

Let us take a look at ten problems that can be easily solved using a multimeter.

When you are doubtful about an electrical outlet

A multimeter can check power output from an outlet. All you have to do is set the meter to 400V AC, place the probes at the combinations given in Table I and check for the respective readings.

TABLE I
CHECKING AN ELECTRICAL OUTLET

Probe 1	Probe 2	Output voltage
Hot	Neutral	230V
Hot	Ground	230V
Neutral	Ground	Zero



Fig. 1: Checking an electrical outlet



Fig. 2: Checking a car battery

TABLE II
CHECKING A CAR BATTERY

Voltage readings	State of charge
12.6V	100 per cent
12.45V	75 per cent
12.24V	50 per cent
12.06V	25 per cent
11.89V	Zero per cent

Note. These readings are at 80°F. Battery voltage readings will drop with temperature, roughly 0.01 volts for every 10°F

When your car is too lazy to start

In the morning, when you start your car, sometimes the motor runs too slow or does not start at all. The first thing you should check is the car battery. The easiest thing that you can do at that moment is pick up a multimeter, set it to 15V DC or more. Then, turn the car lights and ignition off, and touch the meter probes on the terminal posts of the car battery. The readings will be as given in Table II.

If your battery voltage is less than 12.45 volts (75 per cent charged), it is low and should be recharged.

Lot of batteries lying around, but which one is still alive

You can test common household dry-cell batteries with a multimeter. For this, the selector should be set for proper range on DC scale. Household batteries like AA and